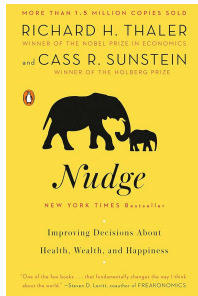


Decision and Risk Analysis

Nudge, Influencing Decisions

DAEN 427/ISEN 427 Fall 2025
Alexander Abuabara





Decision and Risk Analysis

1. Modeling Decisions
2. Modeling Uncertainty
3. Modeling Preferences

Motivation

Nudge

Common Types of Nudges

Applications

Ethics, Critiques, Limits

Designing a Nudge

- ▶ A university engineering building struggles with high energy consumption because many students forget to turn off lights and equipment when leaving laboratories.
- ▶ Traditional reminders such as posters and email announcements have had little effect.
- ▶ The administration wants a design-based solution that encourages people to conserve energy without enforcing strict rules or penalties.

- ▶ Question: As engineers, how could you redesign the physical or digital environment of the building to naturally guide users toward switching off lights and equipment when not in use?
 - ▶ Install motion-activated lighting systems that turn off automatically when no movement is detected after a delay.
 - ▶ Add a small indicator light or display that shows energy saved when lights turn off — giving users instant positive feedback.
 - ▶ Use bright colors or lighting around the switch that changes when the device is left on.
 - ▶ Place switches near the exit so turning them off becomes part of the leaving routine.

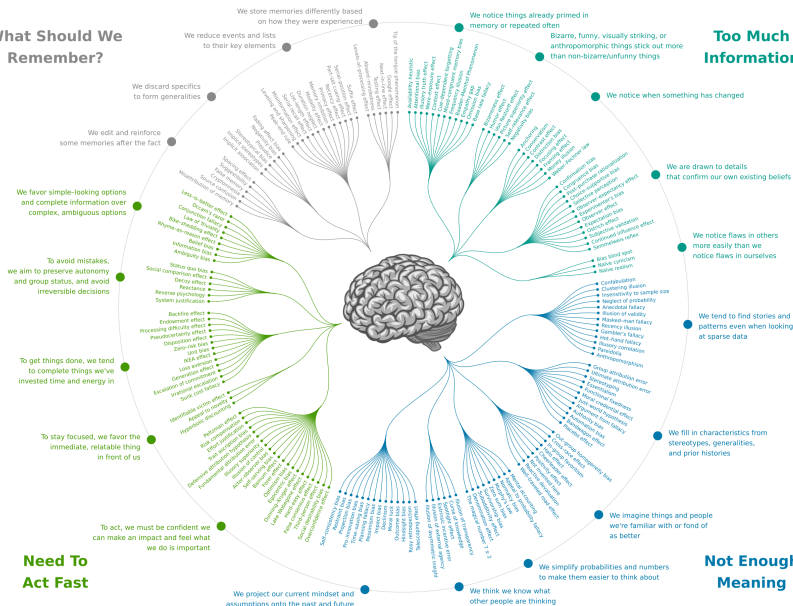
Nudge in one paragraph

... people don't always make the best decisions for themselves bc of **flaws in our psychology**. However, by making **small changes to the environment where decisions are made**, we can gently push people to make **better choices**.

THE COGNITIVE BIAS CODEX

What Should We Remember?

Too Much Information



The Concept of a Nudge

- ▶ A nudge is a small change in the environment that influences the choices people make without restricting their freedom.
- ▶ For example, in a supermarket, products placed at eye level tend to sell more because they're easier to see and reach — illustrating how small environmental tweaks can shape decisions.

You Have Been Nudged



Shoppers are often nudged to choose one product over another (illustration). Photo: Orel Cohen

Under pressure: Nudging increases healthy food choice in a supermarket

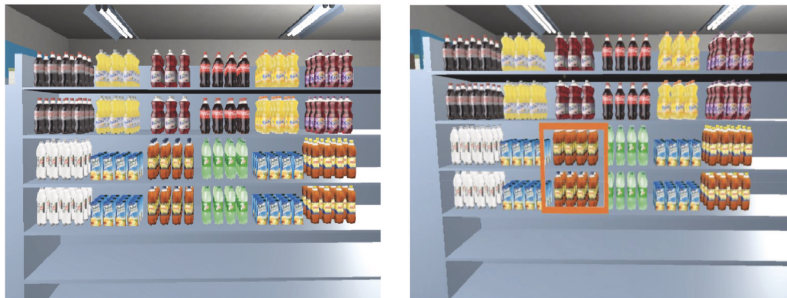


Fig. 1. Example of shelves without (left) and shelves with (right) a salience nudge used in the current study.

Why Talk About Nudging?

- ▶ People often make choices that are not in their long-term interest (health, savings, environment).
- ▶ Policymakers and organizations want to improve choices without banning options.
- ▶ Nudging: steer behavior in a predictable way while preserving freedom of choice.

Behavioral Economics Roots

- ▶ Classical model: fully rational, stable preferences.
- ▶ Reality: heuristics, biases, limited attention.
- ▶ Nudging builds on these predictable departures from rationality.

Three Psychological Biases

1. Anchoring

- ▶ People rely too heavily on the first piece of information they receive. Heuristics, biases, limited attention.
- ▶ Example: In a fundraising study, participants asked to donate \$400 gave an average of \$143, while those asked to donate \$5 gave only \$20. The initial suggestion “anchored” their perception of what was reasonable.



Three Psychological Biases

2. Availability Bias

- ▶ We judge the likelihood of events based on how easily examples come to mind.
- ▶ Example: After an earthquake, sales of earthquake insurance spike — not because quakes are more likely, but because the event is fresh in memory. I still don't have a reference here :(

Three Psychological Biases

3. Representativeness Heuristic

- ▶ We assume things that fit our stereotypes are more common.
- ▶ Example: "Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations."

Which is more probable?

- Linda is a bank teller.
- Linda is a bank teller and is active in the feminist movement.



Three Psychological Biases

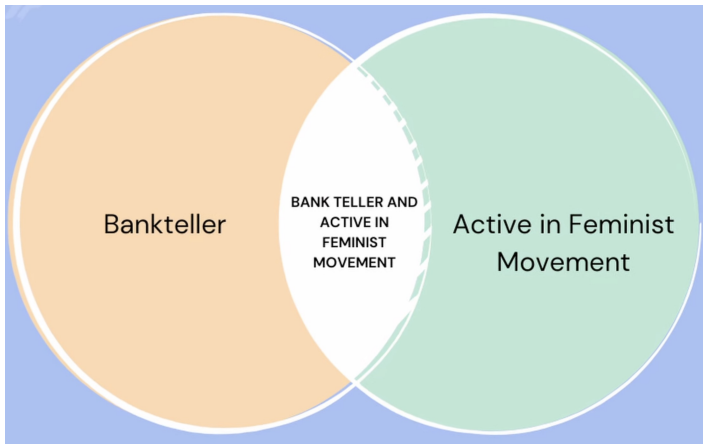
3. Representativeness Heuristic

- ▶ We assume things that fit our stereotypes are more common.
- ▶ Example: "Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations."

Which is more probable?

- Linda is a bank teller.
- Linda is a bank teller and is active in the feminist movement.

Most people think she's more likely a bank teller active in the feminist movement, but statistically, that's less probable because it adds an extra condition. This shows how stereotypes skew rational thinking.
(conjunction fallacy)



Definition

Nudge (informal)

A small change in the **choice architecture** that alters people's behaviour in a predictable way **without forbidding** any options or **significantly changing** their economic incentives.

- ▶ Preserves choice.
- ▶ Low-cost, often simple.
- ▶ Relies on psychology rather than force.

What a Nudge Is / Is Not

Is

- ▶ Default settings
- ▶ Reminders / prompts
- ▶ Framing / ordering
- ▶ Social norm messages

Is Not

- ▶ Bans / mandates
- ▶ Large subsidies/taxes
- ▶ Coercion

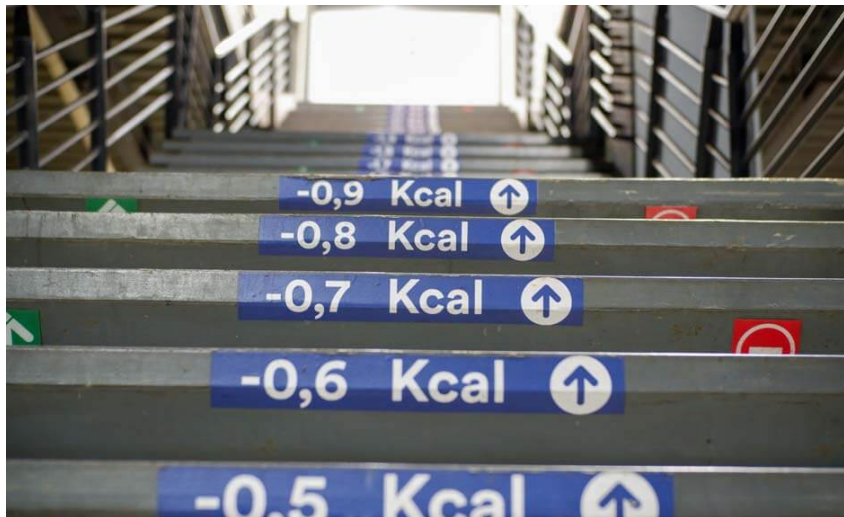
Choice Architecture

- ▶ Choice architecture = the context in which people make decisions.
- ▶ Someone always designs it (form, order, defaults, timing).
- ▶ Nudges are *good* choice architecture.

Key idea

If the environment shapes decisions anyway, we can design it deliberately to promote better outcomes.



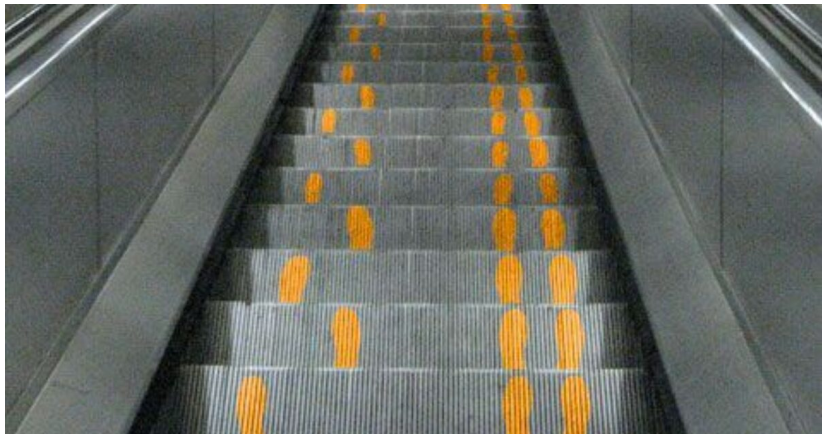






[Click to watch the video on YouTube \(1:47\)](#)



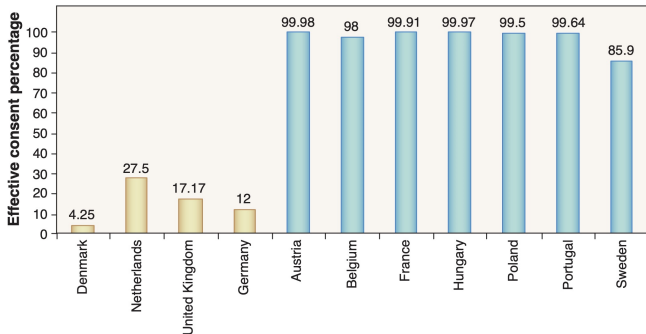


Psychological Levers

- ▶ Status quo / default bias
- ▶ Present bias and limited attention
- ▶ Loss aversion
- ▶ Social influence
- ▶ Salience and simplification

Designing Nudges

- Pre-select an option for people so they're opted in unless they opt out. Countries with opt-out organ donation have much higher donation rates than opt-in countries.

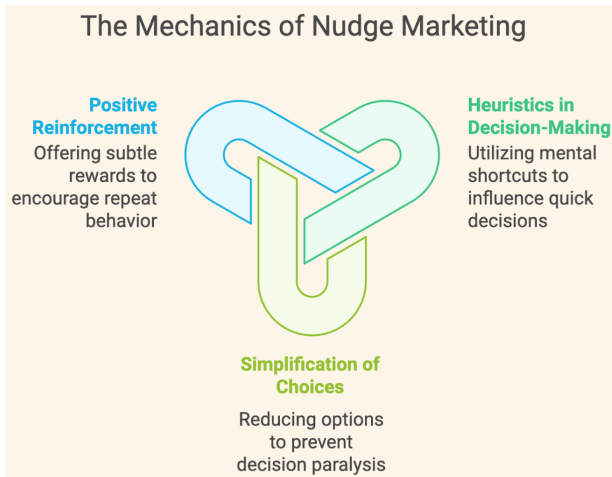


Effective consent rates, by country. Explicit consent (opt-in, gold) and presumed consent (opt-out, blue).

Effective organ donation consent rates of different countries. **"Do Defaults Save Lives?"**, Eric J. Johnson and Daniel Goldstein, Science, 302 (5649), 2001.

Designing Nudges

- Narrow the field. Too many choices can overwhelm people. Instead of showing every song available, Spotify narrows the field by letting users filter by artist, genre, or album — making decisions easier and more satisfying.



1. Defaults

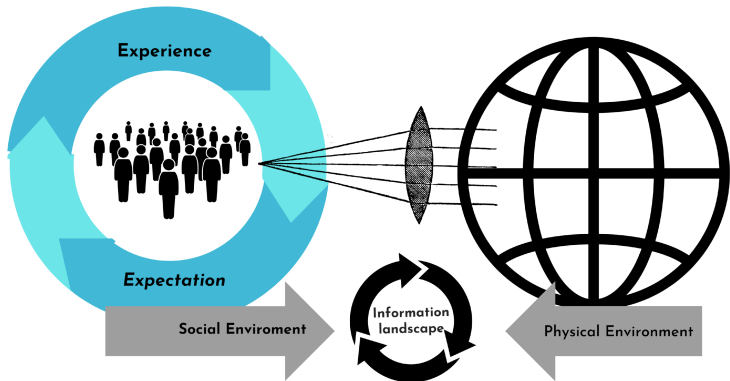
- ▶ People tend to stick with the pre-selected option.
- ▶ Opt-out vs opt-in makes a large difference.
- ▶ Example: automatic enrollment in retirement savings.

2. Framing & Presentation

- ▶ Same information, different framing → different choices.
- ▶ Order of options on a form or website.
- ▶ "Save 90%" vs "10% loss."

3. Social Norms

- ▶ Uses our tendency to conform.
- ▶ "Most people in your area pay on time."
- ▶ "Your energy use is higher than your neighbors."



Social Norms

```
graph TD; A[Social Norms] --> B[Descriptive social norms]; A --> C[Prescriptive social norms]; A --> D[Dynamic social norms]; B --> B1[Describes what people are doing]; B --> B2["X% of people are reducing their meat consumption"]; C --> C1[Conveys what people should be doing]; C --> C2["People should be reducing their meat consumption"]; D --> D1[Illustrates how people's behaviour has been changing]; D --> D2["More and more people are reducing their meat consumption"];
```

Descriptive social norms

Describes what people are doing

"X% of people are reducing their meat consumption"

Prescriptive social norms

Conveys what people should be doing

"People should be reducing their meat consumption"

Dynamic social norms

Illustrates how people's behaviour has been changing

"More and more people are reducing their meat consumption"

4. Reminders & Prompts

- ▶ Inexpensive, scalable.
- ▶ Effective because people forget.
- ▶ SMS/email reminders for appointments, vaccines, payments.

5. Simplification

- ▶ Reduce the number of steps to complete a task.
- ▶ Clear, plain-language instructions.
- ▶ Pre-filled forms.

Public Policy Applications

- ▶ Health: increasing vaccination, healthy food placement.
- ▶ Environment: default green energy plans.
- ▶ Taxes: letters that highlight social norms.
- ▶ Organ donation: opt-out regimes.

Industry & UX

- ▶ Onboarding flows with good defaults.
- ▶ Progress indicators to reduce drop-off.
- ▶ Salient call-to-action buttons.

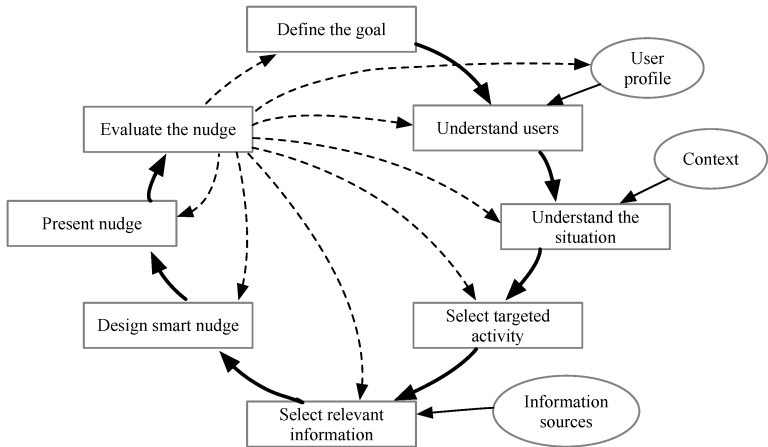
Engineering

Seatbelt Reminder

- ▶ Cars emit a **dinging sound** and show a warning light when the driver's seatbelt is not fastened.
- ▶ Encourages safe behavior without preventing the car from operating.
- ▶ Uses auditory and visual cues as behavioral nudges.

Other Examples:

- ▶ Smart thermostats highlighting eco-friendly settings.
- ▶ Default options in software.
- ▶ Water bottle refill stations.
- ▶ Speed feedback signs.



Is Nudging Ethical?

- ▶ People are always being influenced by how choices are presented — there's no "neutral" way.
- ▶ If influence is inevitable, it should guide people toward better outcomes.
- ▶ They call this approach libertarian paternalism — it preserves freedom of choice while encouraging better decisions.
- ▶ Critics debate this view, but the key idea is: nudges should help, not manipulate.

Criticisms & Risks

- ▶ **Manipulation:** influencing people without them noticing.
- ▶ **Paternalism:** who decides what is "better"?
- ▶ **Context dependence:** may not generalize.
- ▶ **Nudge fatigue:** people stop responding.

Ethical Nudging

- ▶ Transparency: people should be able to understand the nudge.
- ▶ Easy opt-out: preserve autonomy.
- ▶ Proportionate: use nudges for genuine welfare/social goals.
- ▶ Evidence-based: test, evaluate, iterate.

How to Design a Nudge

1. Define the target behaviour (what should change?).
2. Diagnose barriers (forgetting, complexity, lack of salience?).
3. Select the lever (default, reminder, social norm).
4. Prototype and test (A/B testing if digital).
5. Scale the effective version.

- ▶ Humans can be predictably irrational ...
... small changes in context can lead to large behavioral effects.
- ▶ Nudges can help us make better decisions ...
... and still preserve freedom of choice.
- ▶ They work by redesigning environments ...
... grounded in behavioral insights, not removing freedom.
- ▶ Must be evaluated and used ethically.

Discussion

Informational approach

(e.g., health education
and awareness raising)

Nudge

(e.g., behavioral design)

Financial approach

(e.g., incentives and fines)

Regulatory approach

(e.g., law and ordinance)

1. What makes a nudge different from informational, financial, or regulatory approaches in engineering systems?
2. Why might people respond more effectively to subtle design cues than to strict rules or fines?
3. Can you think of examples in your field where a nudge could improve safety, efficiency, or sustainability?
4. What role does human psychology play in designing effective engineering nudges?
5. Compare the cost and practicality of implementing a nudge versus financial or regulatory interventions. Which is more sustainable long-term?
6. How can engineers measure the effectiveness of a nudge compared to traditional approaches?
7. Is there a risk that nudges could manipulate behavior unethically? How can engineers ensure transparency?
8. Should engineers have a moral obligation to use nudges to promote socially desirable outcomes (e.g., safety, environmental protection)?

References

1. Thaler, R.H. & Sunstein, C.R. (2008). *Nudge*.
2. Sunstein, C.R. (2014). *Why Nudge?*
3. Hausman, D. & Welch, B. (2010). *To Nudge or Not to Nudge*.